A novel approach towards drug screening using single cell experiments, isolated heart preparations, multiscale modeling, and machine learning

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Introduction

New drug development
> avg cost: 2.5 billion $
> avg time: 10 years

All drugs, not only cardiac drugs, interact with the ion channels in the heart cell membrane
> up- or down- regulation channel conductivities
> changes in cell-scale action potential
> affects organ-scale cardiac electrophysiology
> potentially lethal arrhythmia risk

Current safety paradigm: blockage of hERG channel (I_{Kr}) and QT prolongation in healthy patients
> High sensitivity
> Low specificity

Goal: combine **multiscale experiments and simulations, high-performance computing, and machine learning** to create a **risk estimator** that can stratify new and existing drugs according to their proarrhythmic potential

Materials and methods

Cell scale simulations

Temporal evolution transmembrane potential:

$$\dot{\phi} = I_{ion}/C_m \quad , \quad \dot{\boldsymbol{q}} = \boldsymbol{g}(\phi, \boldsymbol{q})$$

$$I_{ion} = I_{Kr} + I_{Ks} + I_{K1} + I_{CaL} + I_{Na} + I_{CaNa} + I_{CaK} + I_{Cab} + I_{Nab} + I_{Kb} + I_{to} + I_{NaK} + I_{pCa} + I_{NaCa,i} + I_{NaCa,ss}$$

C_m : membrane capacitance
$I_{ion}(\phi, \boldsymbol{q})$: ionic current
$\boldsymbol{q}$: set of state variables

Drug effects: scaling relevant I_i's by a fractional block

$$I_i^{drug} = [1 - \beta_i] I_i \quad \text{with} \quad \beta_i = \left[1 + [C/IC_{50}]\right]^{-1}$$

Tissue scale simulations

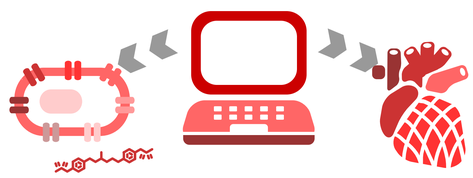
Classic monodomain model, spatiotemporal evolution:

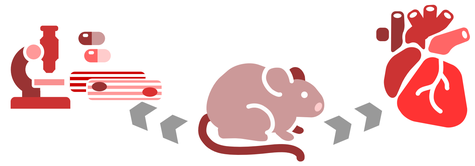
$$\dot{\phi} = div(\boldsymbol{D} \cdot \boldsymbol{\nabla}\phi) - I_{ion}/C_m$$

with $\boldsymbol{D} = D^{\parallel} \boldsymbol{f} \otimes \boldsymbol{f} + D^{\perp} [\boldsymbol{I} - \boldsymbol{f} \otimes \boldsymbol{f}]$

Organ scale simulations

Finite element model, healthy 21yo 50th percentile US male
- Time discretization: 1.0 M equidistant $\Delta t = 0.005ms$
- Space discretization: 6.9M hex elements, edge $0.3mm$
- 7.5M global DOFs and 0.3G local internal variables





Cell scale experiments

Measuring calcium transients in isolated cardiomyocytes
- Male Sprague Dawley rats
- Experimental isolation protocol and cell incubation with intracellular calcium indicator
- Electrical cell stimulation
- Photomultiplier measures cardiomyocyte calcium fluorescence emission

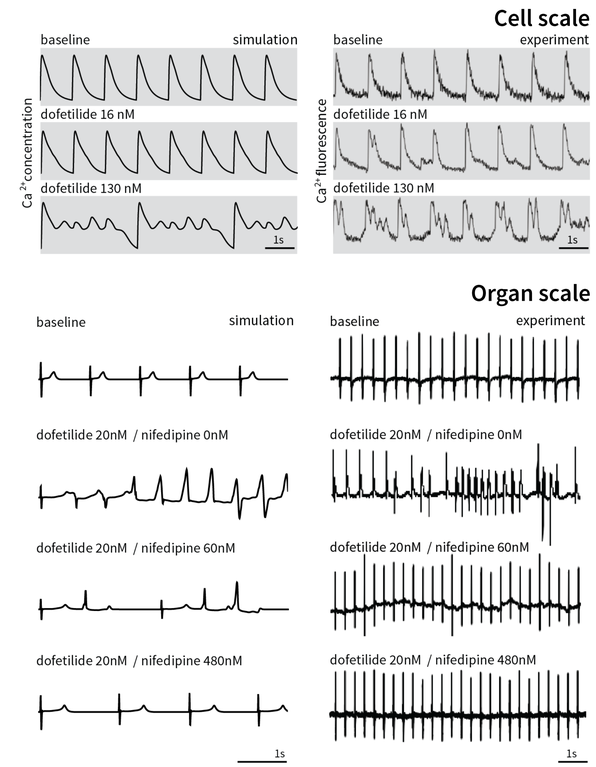
Incubation with different dofetilide concentrations to characterize effect of drugs

Organ scale experiments

Recording ECGs in perfused hearts
- Male Sprague Dawley rats
- Excision, cannulation and connection to constant pressure perfusion Langendorff system
- Electrodes at apex and base

Perfusion with selected concentrations of dofetilide and nifedipine to characterize effect of drugs

Validation

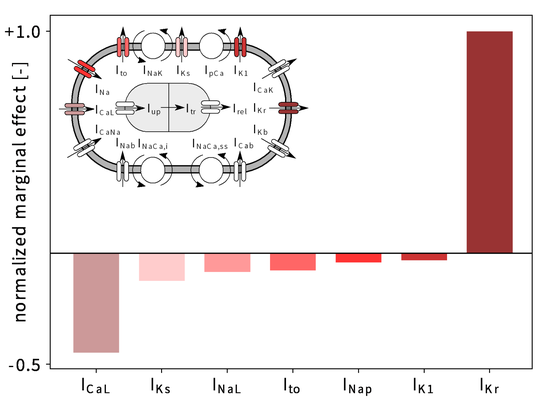


Results and discussion

Ion channels dictate proarrhythmic potential

Early afterdepolarizations (EADs) are **a precursor of TdPs at the cellular level**

Sensitivity analysis
- 500 single mid-wall cell simulations
- Systematically blocking 7 most important ion channels
- Automatic detection of EADs
- Fit logistic regression and quantify marginal effects, a measure that quantifies the effect of each channel blockade on EAD probability

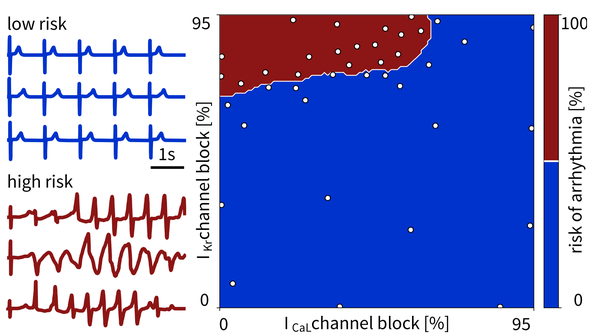


I_{Kr} blockade triggers early afterdepolarizations in simulation and experiment

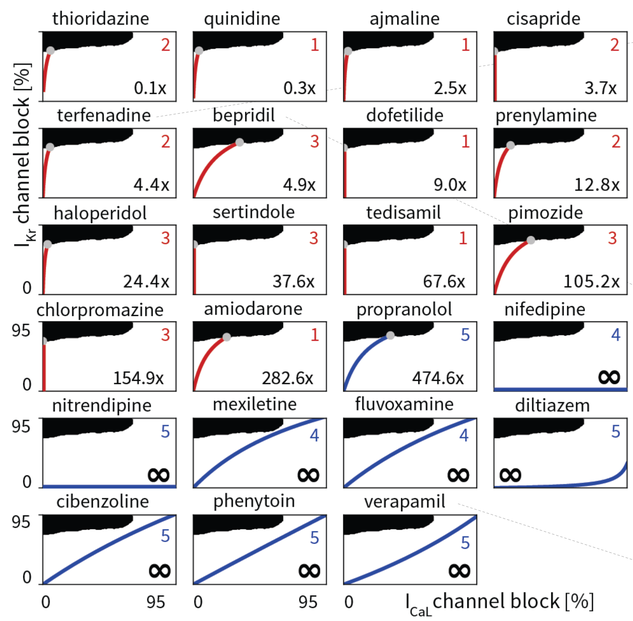
Multi-scale modeling and **machine learning techniques** were used to explore the proarrhythmic risk potential parameter space

- Simulated combined effect of I_{Kr} and I_{CaL} block at different concentrations in multiscale high-fidelity human heart model
- Adopted a particle-learning Gaussian process clarifier to predict an output distribution
- Adaptive sampling: determining point of maximal entropy from previous output distribution to sample new computationally expensive high-fidelity simulation (10 Latin Hypercube start samples and 30 consecutive maximal entropy samples)

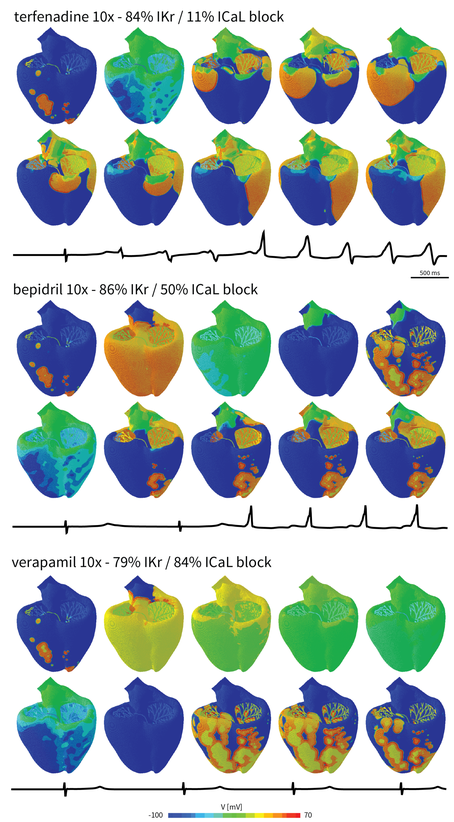
The resulting **proarrhytmic risk classification** divides the parameter space in arrhytmic and nonarrhytmic domains



Risk estimator's critical drug concentrations **align** with **reported torsadogenic risk categories** (top right corners)



This risk stratification is computationally validated for three drugs, where 10x the effective free therapeutic concentration results in arrhythmic patterns for terfenadine and bepidril, but not for verapamil.



- Sahli Costabal F, Seo K, Ashley E, Kuhl E. Classifying drugs by their arrhythmogenic risk using machine learning. Biophys J. 2020; 118:1-12.
- Sahli Costabal F, Matsuno K, Yao J, Perdikaris P, Kuhl E. Machine learning in drug development: Characterizing the effect of 30 drugs on the QT interval using Gaussian process regression, sensitivity analysis, and uncertainty quantification. Comp Meth Appl Mech Eng. 2019; 348:313-333.
- Sahli Costabal F, Yao J, Sher A, Kuhl E. Predicting critical drug concentrations and torsadogenic risk using a multiscale exposure response simulator. Prog Biophys Mol Bio. 2019; 144:61-76.
- Sahli Costabal F, Yao J, Kuhl E. Predicting the cardiac toxicity of drugs using a novel multiscale exposure-response simulator. Comp Meth Biomech Biomed Eng. 2018; 21:232-246.
- Sahli Costabal F, Yao J, Kuhl E. Predicting drug-induced arrhythmias by multiscale modeling. Int J Num Meth Biomed Eng. 2018; 34:e2964.
- Peirlinck M, Sahli Costabal F, Sack KL, Choy JS, Kassab GS, Guccione JM, De Beule M, Segers P, Kuhl E. Using machine learning to characterize heart failure across the scales. Biomech Model Mechanobio. 2019; 18:1987-2001.